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(1) Research Program/Project/Study Title: Socio-economic and Cultural Assessment of the Communities along the Riverine of Roxas City, Capiz	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	
(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center

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☐ Poster		☐ LRDC Livestock Research Development Center
		☐ FFRDC Fisheries and Food Research Development Center
		☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Roxas City
Province:		Capiz
Region:		VI
(5) Clafication		(6) Mode of Implementation
Research:	Development:	☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
☒ Basic	☐ Pilot Testing	
☐ Applied	☐ Tech. Promotion/Commercialization	
(7) Sector/Commodity		(8) Discipline/Field of Specialization
None		Social Science

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(9A) Abstract

This study uses a quantitative and descriptive design to investigate the socio-demographic, economic, and cultural characteristics of the Cadimahan and Palina Riverine communities in Roxas City, Philippines. It employed purposive sampling to select 23 members from Palina and 25 from Cadimahan River tour operators. A survey questionnaire based on the study of the Panay River System was utilized to gather data. Descriptive statistics such as mean, median, and standard deviation were used to summarize and describe the data. Results showed that the two communities have some differences in terms of age, sex, civil status, educational attainment, and religion. However, the overall profiles of the two communities are relatively similar. The Palina riverine Community has a more diverse occupational range and higher daily incomes. Both communities reported implementing environmental conservation practices, however, they face similar environmental issues. It is recommended that further research be conducted and strategies should be implemented to increase the economic standing of the Cadimahan Riverine Community, as well as, increase focus on implementing physical barriers to control erosion on river banks. Additionally, local government should promote and raise awareness of solid waste management and both riverine communities should collaborate to address their shared environmental issues.

(9B) Keywords

Roxas City, Capiz, Riverine, Socio-economic, Cultural Assessment

(10) Introduction

Riverine communities are those that are located along rivers or streams and often rely on these water sources for their livelihoods and way of life. The economic status and cultural practices of these communities can have significant impacts on their challenges and opportunities, and understanding these factors can inform efforts to support and improve the quality of life for these communities.

The economic status and living conditions of riverine communities in Bangladesh and found that these communities often had low levels of education and limited access to healthcare and other basic amenities. Many residents were engaged in subsistence fishing and farming, with low monthly incomes and poor economic prospects (Hossain, 2016).

The occupational diversity and economic status of riverine communities in the Philippines and found that these communities were diverse in terms of their occupations and economic status, with some residents engaged in fishing and agriculture, while others worked in urban centers or engaged in small-scale trade and services. (Gonzalez et al., 2020).

The riverine communities often face significant socio-economic challenges, including low levels of income and education, limited access to healthcare and other basic

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amenities, and high levels of poverty (Gonzalez et al., 2020). These challenges may be influenced by a range of factors, including traditional gender roles and the importance of family and community ties.

In addition to socio-economic factors, cultural and social factors have also been found to impact the effectiveness of environmental conservation and solid waste management practices in developing countries (Nguyen et al., 2017).

The riverine communities have been shown to have strong cultural values and traditions related to the environment, and often engage in conservation practices such as sustainable resource use and reforestation efforts (Rodrigues et al., 2018)

The attitudes and behaviors of urban residents towards environmental issues, such as conservation and waste management, can also vary depending on factors such as education level and personal values (Li et al., 2019). The Cadimahan and Palina Riverine Communities in Roxas City, the attitude towards the preservation, maintenance, and protection of the riverine environment is likely to be an important factor influencing the economic and cultural well-being of these communities. Given the significant socio-economic and cultural challenges faced by riverine communities, it is important to understand the factors that contribute to these challenges in order to inform the development of effective intervention programs. By examining the economic status and cultural practices of the Cadimahan and Palina Riverine Communities in Roxas City, this study aims to provide insight into the challenges and opportunities faced by these communities.

This study was conducted to determine the socio-economic status and cultural assessment of the Cadimahan and Palina Riverine Communities of Roxas City to provide needed intervention programs for the riverine communities.

Specifically, this study sought to answer the following questions:

1. What is the profile of the respondents of Cadimahan and Palina Riverine Communities of Roxas City in terms of age, sex, civil status, highest educational attainment, and religion?
2. What is the economic status of the Cadimahan and Palina Riverine Communities of Roxas City in terms of occupation, daily income, and economic activities?
3. What is the cultural assessment of the Cadimahan and Palina Riverine Communities of Roxas City in terms of environmental conservation practices and solid waste management awareness?
4. What are the problems, issues, and concerns affecting the Cadimahan and Palina Riverine Communities of Roxas City?
5. What intervention programs are necessary to address these issues of Cadimahan and Palina Riverine Communities of Roxas City?

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(11) Materials and Methods or Methodology

Research Design

In this study, the researchers used a quantitative and descriptive design to gather data on the sociodemographic, economic, and cultural characteristics of the Cadimahan and Palina Riverine communities in Roxas City. A descriptive research design is a type of study that is used to describe the characteristics of a population or a phenomenon being studied. It is often used to gather information about a specific group or to understand relationships. This type of design is useful for collecting detailed information about a specific group or for describing the characteristics of a particular phenomenon (Arslan, H., & Ozturk, A, 2019).

The researchers collected data using quantitative methods, such as surveys and questionnaires, and then used descriptive statistics, such as mean, median, and standard deviation, to summarize and describe the data. The data was then further analyzed using these statistical tools (Wang, 2013).

Participants of the Study

The respondents in this study were twenty-three (23) members from the Palina and twenty-five (25) members from Cadimahan river tour operators along the Roxas City, Capiz River. The study used purposive sampling in the identification and selection of the participants. Purposive sampling is a non-probabilistic sampling technique that can be used in descriptive research design. The purpose of purposive sampling is to select a sample that is specifically chosen because the individuals in the sample have some particular characteristic that makes them particularly appropriate or relevant to the research question (Alhassan, 2017).

The criteria for selecting the respondents in the study were narrowed to include only those individuals who: a) have been actively involved as members of Palina and Cadimahan river tour operators for the past five years; b) are fisherfolks and residents of Palina and Cadimahan in Roxas City; c) meet certain characteristics as determined by the researchers' professional judgement and the objective opinions of their colleagues.

Research Instruments

The instruments used in gathering the necessary data for this study was based on the study of the Panay River System was utilized to gather data. It consists of five (5) parts. Part 1 of the questionnaire is the socio-demographic profile of the respondents such as age, sex, civil status, highest educational attainment, and religion.

Part 2 is the socio-economic status such as occupation, monthly income, and economic activities. Part 3 is the cultural assessment such as environmental conservation practices and solid waste management awareness. Part 4 is the the problems, issues, and concerns affecting the Cadimahan and Palina Riverine Communities of Roxas City. Part 5 is the intervention programs are necessary to address these issues of Cadimahan and Palina Riverine Communities of Roxas City.

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The researcher used categorical responses, which involve asking the respondent to choose from a set of predefined categories, and allowing the respondent to check all the choices that apply to them.

Before administering the survey, the researcher pilot-tested the questions with a small sample to ensure that they were clear, unbiased, and well understood. The instruments were validated to ensure its reliability and validity in measuring the construct they claim to assess. The instruments were then pilot tested on at least 30 individuals who were not part of the target respondents of the study to compute their reliability using Cronbach's Alpha (Cheng, 2019).

Data Gathering Procedure

After ensuring the validity and reliability of the research instrument, the researchers obtained permission from the presidents of the PARIDA Association for the Palina Greenbelt River Tour and the LABSFAR Association for the Cadimahan River Tour to administer the questionnaire to residents through purposive sampling. The researcher personally distributed the questionnaires to the selected participants, gave them adequate time to complete them, and subsequently collected them. The data collected was then analyzed using descriptive statistics, such as mean, median, and standard deviation, to summarize and describe the data (Creswell, 2013).

Data Analysis Procedure

The researcher personally gathered and encoded the data that was collected. This data was then processed using statistical measurements and computations, with the help of a statistician, to ensure that the data was analyzed and interpreted accurately (Ribeiro, 2014).

To address problem statements 1, 2, 3, 4, and 5 the mean percentages of scores for every question and component of the entire survey were calculated. The data was analyzed using the Statistical Package for Social Sciences (SPSS) software, which was employed due to the research's descriptive research design. The descriptive statistical tools of SPSS were utilized to provide the most precise values for the analysis and interpretation of the collected data (Creswell, 2013).

Descriptive statistics, such as frequency count and its percentage, mean, and standard deviation, were calculated to make the data more interpretable and to generate a clear understanding of the data that was collected in the study. This helped the researcher to find insights and answer research questions (Mendes, 2012).

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(12) Results and Discussions

Socio-demographic Profile of Respondents

The demographic profiles of Cadimahan and Palina Riverine Communities reveal that they have some distinctions. In terms of age, the majority of respondents in both communities are between 18 and 35 years old, with 44% in Cadimahan and 39.13% in Palina. The proportion of respondents between 36 and 53 years old is also relatively high in both communities, with 32% in Cadimahan and 30.43% in Palina. The percentage of respondents between 54 and 73 years old is 24% in Cadimahan and 26.09% in Palina. The percentage of respondents over 74 years old is 0 in Cadimahan and 4.35% in Palina.

In terms of sex, the majority of respondents are male in both communities, with 68% in Cadimahan and 43.48% in Palina. The proportion of female respondents is 32% in Cadimahan and 56.52% in Palina.

In terms of civil status, the majority of respondents in both communities are married, with 56% in Cadimahan and 39.13% in Palina. The proportion of single respondents is 44% in Cadimahan and 56.52% in Palina. The percentage of widowed and separated respondents is 0 in both communities.

In terms of highest educational attainment, the majority of respondents in both communities have either secondary (Junior High School) or college level of education. In Cadimahan, 44% of respondents have secondary (JHS) education, 24% have college level education, 12% have elementary (Grades 5-6) education, and 4% have either no formal schooling or primary (Grades 1-4) education. In Palina, 34.78% of respondents have secondary (JHS) education, 17.39% have either secondary (SHS) or vocational/technical graduate education, 13.04% have either elementary (Grades 5-6) or primary (Grades 1-4) education, 8.70% have no formal schooling, and 4.35% have college level education.

In terms of religion, the majority of respondents in both communities are Roman Catholic, with 84% in Cadimahan and 84% in Palina. The proportion of Protestant respondents is 12% in Cadimahan and 12% in Palina. The percentage of Iglesia ni Cristo respondents is 4% in Cadimahan and 0 in Palina. The percentage of Seventh Day Adventist and other religion respondents is 0 in both communities.

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Table 1. Profile of the Riverine Communities

Profile of respondents	Cadimahan		Palina	
	f	%	f	%
Age				
18 - 35 year old	11	44.0%	9	39.13%
36 - 53 year old	8	32.0%	7	30.43%
54 - 73 year old	6	24.0%	6	26.09%
74 - 89 year old	0	0.0%	1	4.35%
Total	25	100%	23	100%
Sex				
Male	17	68.0%	13	43.48%
Female	8	32.0%	10	56.52%
Total	25	100%	23	100%
Civil Status				
Single	11	44.0%	13	56.52%
Married	14	56.0%	9	39.13%
Widowed	0	0.0%	0	0.0%
Seperated	0	0.0%	1	4.35%
Total	25	100%	23	100%
Highest Educational Attainment				
No Formal Schooling	1	4.0%	2	8.70%
Primary (Grades 1-4)	1	4.0%	2	8.70%
Elementary (Grades 5-6)	3	12.0%	3	13.04%
Secondary (JHS Grades 7-10)	11	44.0%	8	34.78%
Secondary (SHS Grades 11-12)	0	0.0%	4	17.39%
College Level	6	24.0%	1	4.35%
Voc/Tech Graduate	0	0.0%	3	13.04%
College Graduate	3	12.0%	0	0.0%
Total	25	100%	23	100%
Religion				
Roman Catholic	21	84.0%	21	84%
Protestant	3	12.0%	2	12%
Iglesia ni Cristo	1	4.0%	0	4%
Seventh Day Adventist	0	0.0%	0	0.0%
Others	0	0.00%	0	0.0%
Total	25	100%	23	100%

Socio-Economic Status of Riverine Communities of Roxas City

The Cadimahan and Palina Riverine Communities have drastically different economic standings. In terms of occupation, the majority of the Cadimahan Riverine Community are involved in river tour operation (48%) while the Palina Riverine Community is more diverse with the majority (60.87%) involved in river tour operation and the rest involved in farming (17.39%), fishing (8.70%), and livestock raising (4.35%).

In terms of daily income, the Cadimahan Riverine Community has a more even distribution of income with the majority (40%) earning between P100 and P200 and 28% earning less than P100 per day. On the other hand, the Palina Riverine Community had a much more diverse range of income with 43.48% earning more than P500 and the rest earning between P100 and P400.

In terms of other economic activities, both communities are mostly involved in fishing (13% for Cadimahan and 43.48% for Palina). The Palina Riverine Community is also

heavily involved in agriculture (26.09%) while the Cadimahan Riverine Community is involved in both agriculture (5%) and trade (6%).

Table 2. Economic Status of the Riverine Communities

Components	Cadimahan		Palina	
	f	%	f	%
Occupation				
Carpenter	1	4.0%	2	8.70%
Farming	1	4.0%	4	17.39%
Fishing	10	40.0%	2	8.70%
Livestock raising	1	4.0%	1	4.35%
River Tour Operator	12	48.0	14	60.87%
Total	25	100%	23	100%
Daily Income				
Below P100	7	28.0%	1	4.35%
P 100 – P200	10	40.0%	5	21.74%
P 201 – P300	1	4.0%	2	8.70%
P 301 – P400	4	16.0%	2	8.70%
P404 – P500	2	8.0%	3	13.04%
P500 above	1	4.0%	10	43.48%
Total	25	100%	23	100%
Other Economic Activities				
Agriculture	5	5.0%	6	26.09%
Forestry	1	1.0%	1	4.35%
Fishing	13	13.0%	10	43.48%
Quarrying	0	0.0%	0	0.0%
Trade	6	6.0%	6	26.09%
Total	25	100%	23	100%

Environmental Conservation Practices

The Cadimahan and Palina Riverine Communities have similar levels of environmental conservation practices. In Cadimahan, 64% of respondents reported that they plowed their farms, compared to 56.2% in Palina. Similarly, 80% of Cadimahan respondents reported using agroforestry farming systems, compared to 86.96% in Palina. The majority of respondents in both communities reported using water conservation and management practices, compost, biodegradable products, natural alternatives to chemical pesticides and fertilizers, and tree planting. The highest level of agreement was in the use of best management practices to reduce agricultural runoff into rivers, with 96% of Cadimahan respondents and 95.65% of Palina respondents reporting that they implemented these practices. The lowest level of agreement was in the use of physical barriers to control erosion on river banks, with 88% of Cadimahan respondents and 89.96% of Palina respondents reporting that they implemented this practice.

The Cadimahan and Palina Riverine Communities have similar levels of environmental conservation practices. The majority of respondents in both communities reported that they implemented practices to reduce water waste and pollution, use compost and natural alternatives to chemical pesticides and fertilizers, and plant trees. The highest level of agreement was in the use of best management practices to reduce agricultural runoff into rivers, and the lowest level of agreement was in the use of physical barriers to control erosion on river banks. However, even with the lowest level of agreement, a majority of respondents in both communities reported that they implemented this practice.

Table 3. Environmental Conservation Practices

Statement	Cadimahan		Palina	
	Yes No. (%)	No No. (%)	Yes No. (%)	No No. (%)
1. Do you plow your farm?	16 64.0%	9 36.0%	13 56.20%	10 43.48%
2. Do you apply an agroforestry farming system on your farm?	20 80.0%	5 20.0%	20 86.96%	3 13.04%
3. Do you implement water conservation and management practices to reduce water waste and pollution?	22 88.0%	3 12.0%	19 82.61%	4 17.39%
4. Do you apply compost to your crop?	20 80.0%	5 20.0%	18 78.26%	5 21.74%
5. Do you use biodegradable products in activities near rivers?	18 72.0%	7 28.0%	16 69.57%	7 30.43%
6. Do you reduce the use of chemical pesticides and fertilizers, and replace them with natural alternatives such as compost, manure, and mulch?	22 88.0%	3 12.0%	20 89.96%	3 13.04%
7. Did you plant trees in your home/farmland/surrounding areas?	21 84.0%	4 16.0%	21 91.30%	2 8.70%
8. Do you utilize no-till or low-till farming techniques to minimize soil disruption and reduce erosion?	21 84.0%	4 16.0%	19 82.61%	4 17.39%
9. Do you implement best management practices to reduce agricultural runoff into rivers?	24 96.0%	1 4.0%	22 95.65%	1 4.35%
10. Do you use physical barriers to control erosion on river banks?	22 88.0%	3 12.0%	20 89.96%	3 13.04%

Solid Waste Management Awareness of Riverine Communities

The Cadimahan and Palina Riverine Communities have a high level of awareness of solid waste management. The majority of respondents in both communities answered 'Yes' to all of the statements, with nearly 100% of respondents in Cadimahan and 95-100% of respondents in Palina answering 'Yes'. The only exception was statement 2, where 80% of respondents in Cadimahan and 95.65% of respondents in Palina answered 'Yes'. This indicates that both communities are aware of the importance of solid waste management and the need to separate biodegradable waste from non-biodegradable waste. They also demonstrate a willingness to reuse materials, recycle waste materials, reduce household waste, and observe 'No Plastic Days' of the local government unit. Finally, both communities understand the environmental and health impacts of solid waste improperly disposed of in the Riverine of Roxas City, including siltation, pollution, flooding, overflowing, and the death of aquatic plants and species.

Table 4. Solid Waste Management Awareness

Statement	Cadimahan		Palina	
	Yes No. (%)	No No. (%)	Yes No. (%)	No No. (%)
1. I collect household solid waste for proper disposal.	23 92.0%	2 8.0%	19 82.61%	4 17.39%
2. I separated biodegradable from non-biodegradable.	20 80.0%	5 20.0%	22 95.65%	1 4.35%
3. I reuse materials such as craft boards, plastic bags, cans, bottles, etc.	19 76.0%	6 24.0%	22 95.65%	1 4.35%
4. I believe that proper waste disposal is necessary for the environment.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
5. I recycle waste materials in other forms.	18 72.0%	7 28.0%	23 100.0%	0 0.0%
6. I reduce my household waste.	15 60.0%	10 40.0%	20 86.96%	3 13.04%
7. I observe "No Plastic Days" of the local government unit.	25 100.0%	0 0.0%	19 82.61%	4 17.39%
8. When solid waste is thrown/dumped into the Riverine of Roxas City, I know it can silt and pollute the river.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
9. When solid waste is thrown/dumped into the Riverine of Roxas City, I know it can cause flooding and overflowing of river banks.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
10. When solid waste is thrown/dumped into the Riverine of Roxas City, I know it can kill aquatic plants and species	25 100.0%	0 0.0%	23 100.0%	0 0.0%

Problems, Issues, and Concerns affecting the Cadimahan and Palina Riverine Communities of Roxas City

Cadimahan and Palina Riverine Communities of Roxas City are experiencing a variety of issues, problems, and concerns. The most pressing issue is that fish populations are declining, with 76% of Cadimahan respondents and 69.57% of Palina respondents saying it is a problem. Biodiversity is also not thriving, with 60% of Cadimahan respondents and 56.52% of Palina respondents saying it is a problem. Kaingin and cutting of trees is also a major issue, with 84% of Palina respondents saying it is a problem. Polluted river water and flooding due to siltation are also issues, with 40% of Cadimahan respondents and 43.48% of Palina respondents saying it is a problem. Finally, high river levels were cited as a problem by 60% of Cadimahan respondents and 56.52% of Palina respondents.

The issues, problems, and concerns affecting the Cadimahan and Palina Riverine Communities of Roxas City are largely similar. Both communities are facing similar environmental issues, including declines in fish populations, biodiversity, and polluted river water, as well as kaingin and cutting of trees, flooding due to siltation, and high river levels.

Table 5. Problems, Issues, and Concerns affecting the Cadimahan and Palina Riverine Communities of Roxas City

Statement	Cadimahan		Palina	
	Yes No. (%)	No No. (%)	Yes No. (%)	No No. (%)
1. Fish populations are declining.	19 76.0%	6 24.0%	15 69.57%	8 34.78%
2. Biodiversity is not thriving.	15 60.0%	10 40.0%	13 56.52%	10 43.48%
3. Kaingin and cutting of trees.	4 16.0%	21 84.0%	13 56.52%	10 43.48%
4. Polluted River water.	10 40.0%	15 60.0%	15 65.22%	8 34.78%
5. Flooding due to siltation.	10 40.0%	15 60.0%	13 56.52%	10 43.48%
6. High River level.	10 40.0%	15 60.0%	10 43.48%	13 56.52%
Others	0 0.0%	0 0.0%	0 0.0%	0 0.0%

Intervention programs are necessary to address these issues of Cadimahan and Palina Riverine Communities of Roxas City

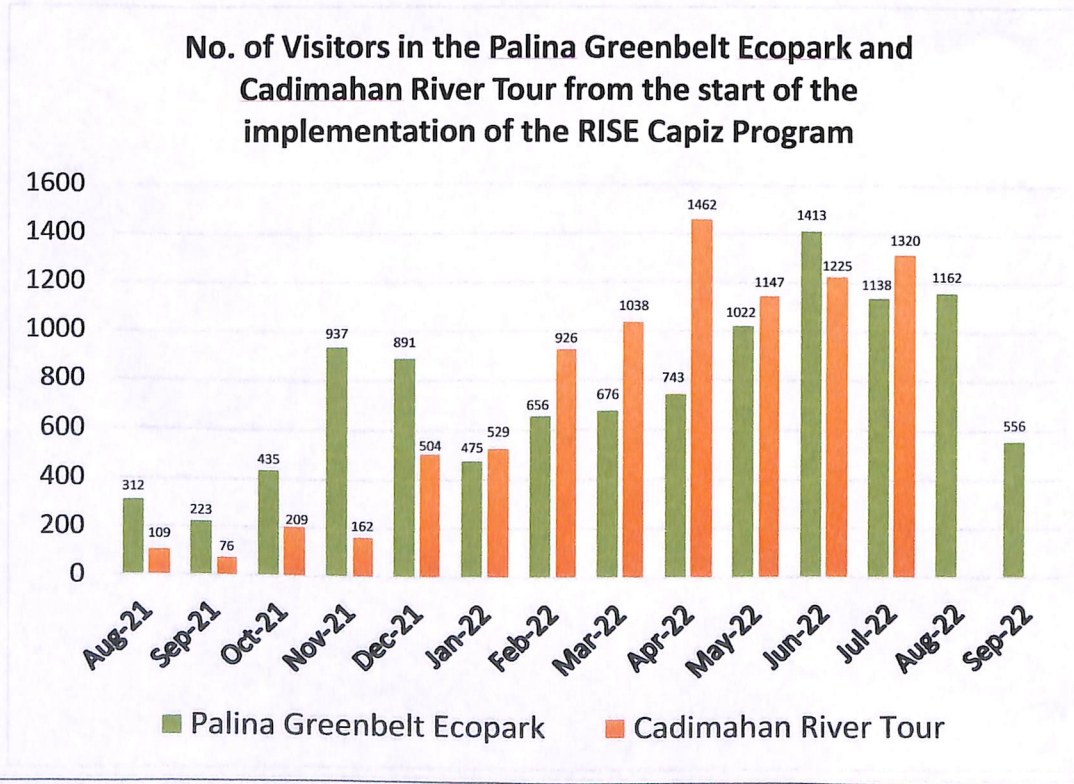
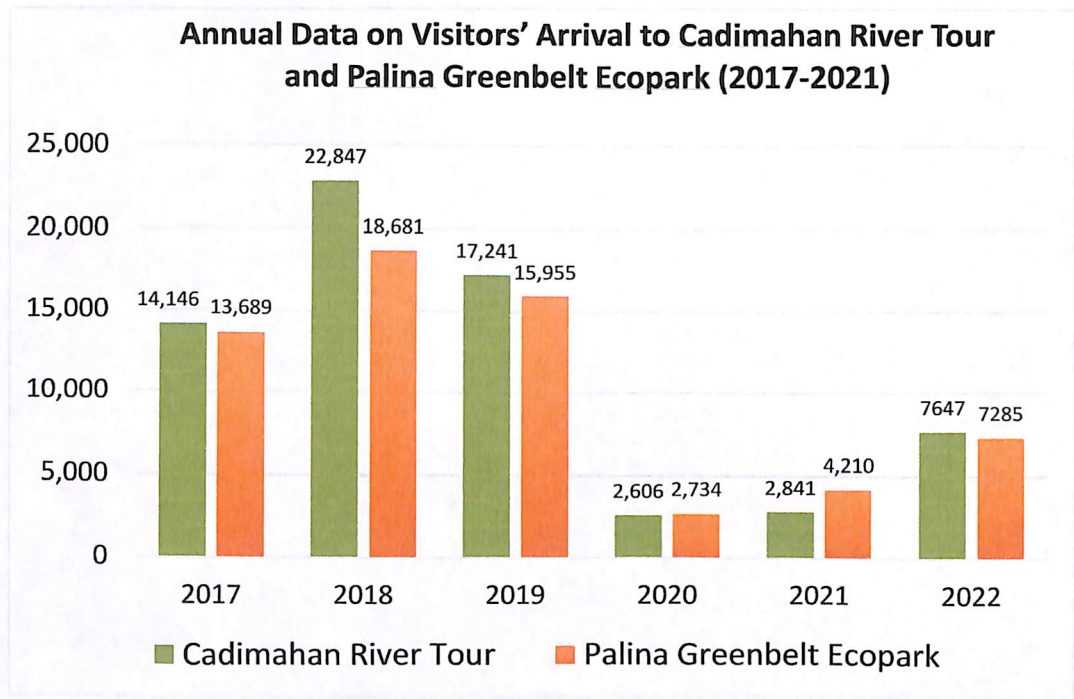
The Cadimahan and Palina Riverine Communities of Roxas City strongly support the need for intervention programs to address the issues in the area, with 100% of the respondents in Cadimahan and 86.96% in Palina responding positively. Furthermore, all respondents from Cadimahan and 100% from Palina agreed that protected areas should be established and natural habitats should be restored, sustainable fishing practices should be implemented, resources should be used responsibly, kaingin and unregulated logging should be reduced or banned, and cleanup drives should be organized. However, only 80% of respondents from Cadimahan and 65.22% of respondents from Palina agreed that wastewater should be treated and filtered before it is discharged into rivers, and only 100% of respondents from Cadimahan and 34.78% of respondents from Palina agreed that riverbanks and dikes should be reinforced to prevent flooding.

It suggests that residents of Cadimahan and Palina Riverine Communities of Roxas City recognize the need for intervention programs and are open to taking action to address the issues in the area. It is important that the relevant authorities take appropriate steps to implement these interventions and ensure that the necessary resources are allocated for their implementation.

Table 6. Intervention programs for the Cadimahan and Palina Riverine Communities of Roxas City

Statement	Cadimahan		Palina	
	Yes No. (%)	No No. (%)	Yes No. (%)	No No. (%)
1. Establish protected areas and restore natural habitats.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
2. Implement sustainable fishing practices.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
3. Promote responsible use of resources.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
4. Reduce or ban kaingin and unregulated logging.	25 100.0%	0 0.0%	23 100.0%	0 0.0%
5. Treat and filter wastewater before it is discharged into rivers.	20 80.0%	5 20.0%	20 86.96%	3 13.04%
6. Reinforce riverbanks and dikes to prevent flooding.	25 100.0%	0 0.0%	15 65.22%	8 34.78%
7. Cleanup drive.	25 100.0%	0 0.0%	23 100.0%	0 0.0%

Table 7. As to the annual tourist visited at Cadimahan River Tour and Palina Greenbelt Ecopark.



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(13) Conclusions and Recommendations

Conclusions

The results of the socio-demographic profile of the respondents from Cadimahan and Palina Riverine Communities show that the two communities have some differences in terms of age, sex, civil status, educational attainment, and religion. However, the overall profiles of the two communities are relatively similar. The economic standing of the riverine communities differs significantly from each other, with the Palina riverine Community having a more diverse occupational range and higher daily incomes. They have similar levels of environmental conservation practices and generally reported implementing practices to reduce water waste and pollution, use compost and natural alternatives to chemical pesticides and fertilizers, and plant trees. Both riverine communities have a high level of awareness of solid waste management however facing similar environmental issues. Interventions are necessary to address the issues of the riverine communities of Roxas City. These include establishing protected areas and restoring natural habitats, implementing sustainable fishing practices, using resources responsibly, reducing or banning kaingin and unregulated logging, organizing cleanup drives, treating and filtering wastewater prior to discharge into rivers, and reinforcing riverbanks and dikes to prevent flooding.

Recommendations

Based on the drawn conclusions, it is recommended that further research be conducted to understand how these differences affect the communities in terms of livelihood, health, and other socio-economic issues. The strategies should be implemented to increase the economic standing of the Cadimahan Riverine Community, such as providing access to more diverse employment opportunities and increasing daily incomes. There should be an increased focus on implementing physical barriers to control erosion on river banks to further reduce water pollution in the Cadimahan and Palina Riverine Communities. The local government unit continues to promote and raise awareness of solid waste management by providing more education and awareness campaigns, as well as enforcing existing regulations and policies. Both riverine communities of Roxas City should collaborate to address their shared environmental issues. The relevant authorities must take appropriate steps to implement these interventions and ensure that the necessary resources are allocated for their implementation.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
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(16) Economic Analysis

(This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI), and payback period.

(17) Acknowledgments

(Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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